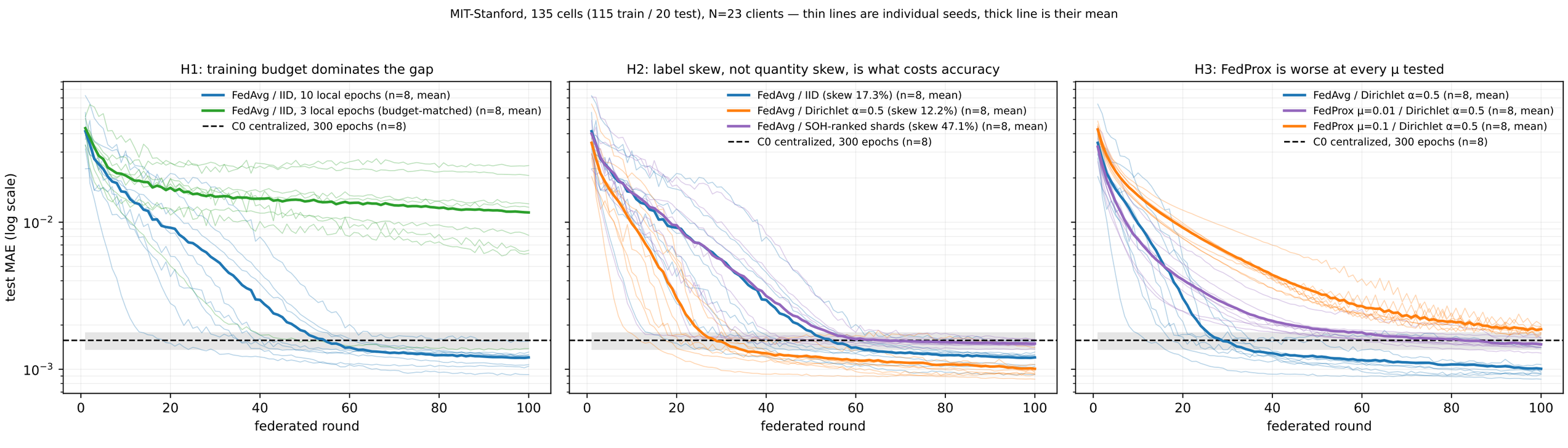

MIT-Stanford, 135 cells (115 train / 20 test), N=23 clients — thin lines are individual seeds, thick line is their mean
H1: training budget dominates the gap
FedAvg / IID, 10 local epochs (n=8, mean)
FedAvg / IID, 3 local epochs (budget-matched) (n=8, mean)
C0 centralized, 300 epochs (n=8)
test MAE (log scale)
10^{-2}
10^{-3}
0
20
40
60
80
100
federated round
H2: label skew, not quantity skew, is what costs accuracy
FedAvg / IID (skew 17.3%) (n=8, mean)
FedAvg / Dirichlet α=0.5 (skew 12.2%) (n=8, mean)
FedAvg / SOH-ranked shards (skew 47.1%) (n=8, mean)
C0 centralized, 300 epochs (n=8)
0
20
40
60
80
100
federated round
H3: FedProx is worse at every μ tested
FedAvg / Dirichlet α=0.5 (n=8, mean)
FedProx μ=0.01 / Dirichlet α=0.5 (n=8, mean)
FedProx μ=0.1 / Dirichlet α=0.5 (n=8, mean)
C0 centralized, 300 epochs (n=8)
0
20
40
60
80
100
federated round